

SHEN ZEHAO

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EDUCATION

Zhejiang University

B.E. in Information Engineering

College of Information Science and Electronic Engineering

Overall GPA: 4.09/4.3

Sept. 2023 – Jun. 2027 (Expected)

RESEARCH & INNOVATION EXPERIENCE

Summer Research – LLM Agent Architectures

Nanyang Technological University, Singapore

Jul. 2026 – Present

Advisor: Prof. Chau Yuen

- Reproduced and analyzed **WirelessAgent++**, an automated agentic workflow framework for wireless tasks, including modular agent operators, **MCTS-based workflow search**, feedback mechanisms, and evaluation on WirelessBench.
- Investigated improvements to agent workflows through **tool use**, **execution feedback**, **iterative repair**, and **work-flow optimization**, while establishing reproducible baselines and resolving implementation and evaluation issues in the open-source framework.
- Developing a **taxonomy and design space for LLM reasoning architectures**, studying how task characteristics and available feedback determine the effectiveness of mechanisms such as code execution, self-refinement, search, and automated workflow design, toward an **architecture selection framework**.

Telecom AI Dataset Construction

Zhejiang University

March 2026 – June 2026

Advisor: Dr. Chongwen Huang

- Contributed to the construction of reasoning datasets for **TelecomGPT-R1**, focusing on mathematical derivation and numerical computation problems in telecommunications.
- Worked on data-processing pipelines that transform textbooks, lecture notes, papers, and open problem sets into self-contained **problem–reasoning–answer** samples, including problem filtering, decomposition, reasoning-chain generation, and sample augmentation.
- Participated in multi-stage sample verification using numerical consistency checks, unit-aware comparison, symbolic equivalence, **Python** recomputation, and independent re-solving to improve data correctness and reproducibility.

EDU-Mate: Multimodal Intelligent Learning Assistant

March 2026 – July 2026

- Developed a multimodal learning assistant integrating streaming speech recognition, visual understanding, structured notes, knowledge graphs, RAG, and LLM-based Agent workflows.
- Designed the core **Agent orchestration** layer and implemented system integration across the **FastAPI** backend, **React/TypeScript** frontend, local inference modules, and real-time WebSocket communication.
- Led system integration and debugging; the project received **Third Prize** in the Intel Cup Embedded System Design Contest under the National Undergraduate Electronic Design Contest.

ACADEMIC PROJECTS

5-Stage Pipelined RISC-V CPU

October 2025 – December 2025

- Architected a 32-bit, 5-stage pipelined RISC-V microprocessor from scratch using **Verilog HDL**, successfully synthesizing and deploying it on a **Nexys Video FPGA**.
- Engineered a **Forwarding Unit** and **Hazard Detection Unit** to resolve Read-After-Write (RAW) data hazards and Load-Use stalls, optimizing pipeline efficiency.

- Validated datapath routing and control logic correctness through comprehensive module-level and top-level testbenches in **ModelSim**.

Temperature Control System Design

June 2025 – July 2025

- Designed the hardware schematic and **PCB**, integrating high-precision sensors (I2C GY906, NTC) and a **MOSFET-based PWM** drive circuit for a 10W heating element.
- Implemented a custom **PID control algorithm** on **STM32**, utilizing adaptive parameters and sliding window filtering to achieve precise temperature regulation.
- Executed hardware-software joint debugging, performing **oscilloscope signal verification** and NTC curve fitting to ensure robust system reliability.

Remote Sound Control System

January 2026

- Engineered an end-to-end voice control simulation platform in **MATLAB**, implementing **LPC** to achieve a 6.3:1 data compression ratio over an **AWGN** channel.
- Developed a robust speech recognition pipeline using adaptive **VAD** and **MFCC-CMVN** feature extraction, enhancing pattern matching via a custom length-penalized **DTW algorithm** to ensure high accuracy at 0-dB SNR.
- Formulated a **state-space model** for a DC motor and deployed a **Finite State Machine (FSM)** to translate continuous voice command streams into real-time, incremental kinematic control.

TECHNICAL STRENGTHS

Programming	C/C++, Python, MATLAB, Verilog
Language	TOEFL after January 21, 2026: 5.0 (R:6, L:6, W:4, S:4.5), Mandarin (Native)

HONORS & AWARDS

Third Prize, Intel Cup Embedded System Design Contest , National Undergraduate Electronic Design Contest	<i>Jul. 2026</i>
Zhejiang Provincial Government Scholarship , Zhejiang Provincial Government	<i>Nov. 2025</i>
Third-Class Scholarship of National Talent Training Base , Zhejiang University	<i>Nov. 2025</i>
Outstanding Student for Academic Excellence , Zhejiang University (Awarded twice)	<i>Oct. 2024 & 2025</i>
Second Prize in Provincial Physics Innovation Competition , Zhejiang Physical Society	<i>Dec. 2024</i>